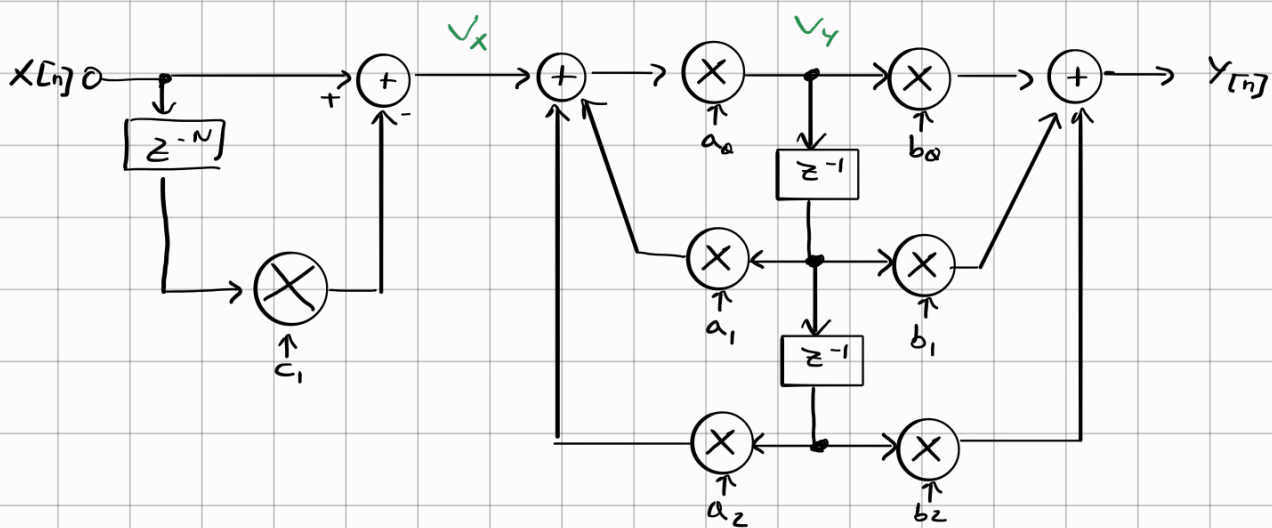




1)

$$Y[n] = (b_0 + b_1 z^{-1} + b_2 z^{-2}) V_y$$

$$V_y = a_0 V_x + a_0 (a_1 z^{-1} + a_2 z^{-2}) V_y$$

$$\hookrightarrow V_y = \frac{a_0}{1 - a_0 a_1 z^{-1} - a_0 a_2 z^{-2}} V_x$$

$$V_x = X[n] (1 - c_1 z^{-N})$$

$$\Rightarrow Y[n] = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}} (1 - c_1 z^{-N}) X[n]$$

$$\Rightarrow H[n] = (1 - c_1 z^{-N}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

2) $a_0 = 1, a_1 = 1, a_2 = 0, b_0 = \frac{1}{N}, b_1 = 0, b_2 = 0, c_1 = 1, N = \{3, 4, 5\}$

$$H(z) = 1 - c_1 z^{-N} \cdot \frac{\frac{1}{N} + b_1 z^{-1} + b_2 z^{-2}}{1 - z^{-1} - a_2 z^{-2}} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$\Rightarrow \boxed{H(z) = \frac{1}{N} \cdot \frac{1 - z^{-N}}{1 - z^{-1}}}$$

$$H(\omega) = \frac{1}{N} \cdot \frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} = \frac{1}{N} \cdot \frac{e^{-j\omega \frac{N}{2}}}{e^{-j\frac{\omega}{2}}} \cdot \frac{(e^{j\omega \frac{N}{2}} - e^{-j\omega \frac{N}{2}})}{(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}})}$$

$$\boxed{H(\omega) = \frac{1}{N} \cdot \frac{\sin(N\frac{\omega}{2})}{\sin(\frac{\omega}{2})} e^{-j\frac{\omega}{2}(N-1)}}$$

- Es un Filtro FIR ya que la componente del Integrador (IIR) resulta anulada.

- Ventajas: Al tener mayor grado (mayor N) se obtienen más coeficientes y se puede cumplir plantillas más exigentes.

$$x^N - 1 = \begin{cases} (x-1)^N, & N \text{ impar} \\ (x-1)^{\frac{N}{2}}(x+1)^{\frac{N}{2}}, & N \text{ par} \end{cases} \Rightarrow H(z) = \frac{1}{N} \cdot \frac{z^N - 1}{z^{N-1}(z-1)}$$

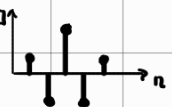
$$H_{\text{imp}}(z) = \frac{1}{N} \cdot \frac{(z-1)^N}{z^{N-1}(z-1)}$$

$$\left\{ \begin{array}{l} H_{\text{imp}}(z) = \frac{1}{N} \cdot \frac{(z-1)^{N-1}}{z^{N-1}} \\ H_{\text{par}}(z) = \frac{1}{N} \cdot \frac{(z-1)^{\frac{N}{2}-1}(z+1)^{\frac{N}{2}}}{z^{N-1}} \end{array} \right.$$

$$\left\{ \begin{array}{l} H_3(z) = \frac{1}{3} \cdot (1 - 2z^{-1} + z^{-2}) \\ \Rightarrow y[n] = \frac{1}{3} \cdot (x[n] - 2x[n-1] + x[n-2]) \\ \\ H_4(z) = \frac{1}{4} \cdot (1 + z^{-1} - z^{-2} - z^{-3}) \\ \Rightarrow y[n] = \frac{1}{4} (x[n] + x[n-1] - x[n-2] - x[n-3]) \\ \\ H_5(z) = \frac{1}{5} (1 - 4z^{-1} + 6z^{-2} - 4z^{-3} + z^{-4}) \\ \Rightarrow y[n] = \frac{1}{5} (x[n] - 4x[n-1] + 6x[n-2] - 4x[n-3] + x[n-4]) \end{array} \right.$$

• Si $x[n] = \delta[n] \Rightarrow y[n] = h[n]$

Para lograr una $h[n] = (1, 1, 1, 1, 1, 1, 1)$ se necesitaría una $H(z)$ del estilo de $H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}$. Con el Filtro de Media Móvil, se necesitaría un $N=7$, pero se debería solucionar el problema del promedio ($k = \frac{1}{N}$) y los coeficientes $\neq 1$. Ej: para $H_5 \rightarrow h_5[n] = (\frac{1}{5}, -\frac{4}{5}, \frac{6}{5}, -\frac{4}{5}, \frac{1}{5})$



$$3) \quad H[z] = (1 - c_1 z^{-N}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 - a_1 z^{-1} - a_2 z^{-2}}$$

$$\bullet \quad Y[n] = \frac{1}{2} (x[n] - x[n-2])$$

$$Y[z] = \frac{1}{2} (X[z] - X[z] z^{-2})$$

$$H[z] = \frac{1}{2} - \frac{1}{2} z^{-2} //$$

$$a_0 = 1, \quad c_1 = a_1, \quad N = 1, \quad a_2 = 0$$

$$b_0 = \frac{1}{2}, \quad b_1 = 0, \quad b_2 = \frac{1}{2}$$

$$4) H(z) = (1 - \alpha_1 z^{-1}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{\alpha_0} - \alpha_1 z^{-1} - \alpha_2 z^{-2}}$$

$$\begin{cases} \alpha_0 = 1 \\ \alpha_1 = \beta = 0,9 \\ b_0 = 1 \\ b_1 = -1 \end{cases} \Rightarrow H(z) = \frac{1 - z^{-1}}{1 - \beta z^{-1}} = \frac{z - 1}{z - \beta}$$

$$o \quad H(\omega) = \frac{e^{j\omega} - 1}{e^{j\omega} - \beta}, \quad H(\omega) \Big|_{\omega=0} = 0 \quad \checkmark$$

$$o \quad H(e^{j\omega}) = \frac{e^{j\omega} - 1}{e^{j\omega} - \beta} = \frac{\cos \omega + j \sin \omega - 1}{\cos \omega + j \sin \omega - \beta}$$

$$|H(e^{j\omega})| = \frac{\sqrt{(\cos^2 \omega + 1 - 2 \cos \omega) + \sin^2 \omega}}{\sqrt{(\cos^2 \omega + \beta^2 - 2\beta \cos \omega) + \sin^2 \omega}}$$

$$|H(e^{j\omega})| = \frac{\sqrt{2 - 2 \cos \omega}}{\sqrt{\beta^2 + 1 - 2\beta \cos \omega}} //$$

$$o \quad 20 \log |H(e^{j\omega})| \Big|_{\omega=\frac{\pi}{10}} = 20 \log |H(e^{j\omega})| \Big|_{\omega=\pi} - 3 \text{ dB}$$

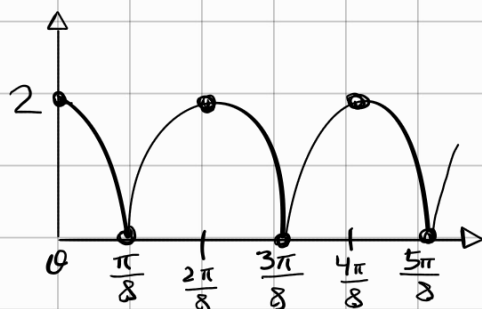
$$20 \log \frac{|H(\omega=\frac{\pi}{10})|}{|H(\omega=\pi)|} = -3 \text{ dB} \rightarrow \frac{|H(\omega=\frac{\pi}{10})|}{|H(\omega=\pi)|} = 10^{-\frac{3}{20}}$$

$$\sqrt{\frac{2(1 - \cos \frac{\pi}{10})}{\beta^2 + 1 - 2\beta \cos \frac{\pi}{10}}} = 10^{-\frac{3}{20}} \rightarrow \boxed{\beta = 0,727}$$

$$5) H[z] = (1 - c_1 z^{-1}) \cdot \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

$$H[z] = 1 + z^{-8} \rightarrow y[n] = 1 + x[n-8] //$$

$$h(\omega) = 1 + e^{-j8\omega} = 2 \cos(4\omega) e^{-j4\omega} //$$



• nulls: $\omega = \frac{\pi}{8} + \frac{\pi}{4}k$

• peaks: $\omega = \frac{\pi}{4}k$